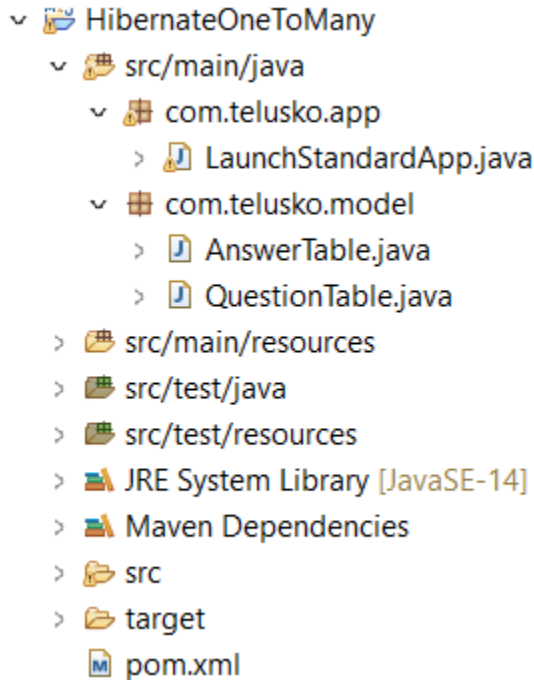


Hibernate One-to-Many & Many-to-One Association

- **Definition:** A one-to-many association refers to a relationship where one instance of an entity can be associated with multiple instances of another entity.
- **Project Structure**



Example in Code:

```
package com.telusko.model;

import java.util.List;

import jakarta.persistence.CascadeType;
import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.OneToOne;
```

```

@Entity
public class QuestionTable
{
    @Id
    @Column(name="question_id")
    private Integer id;

    private String question;

    @OneToMany(cascade=CascadeType.ALL)
    private List<AnswerTable> answerList;

    public QuestionTable()
    {
        System.out.println("Zero Param Constructor of Question");
    }

    @Override
    public String toString() {
        return "QuestionTable [id=" + id + ", question=" + question + ",
answerList=" + answerList + "]";
    }

    public Integer getId() {
        return id;
    }

    public void setId(Integer id) {
        this.id = id;
    }
}

```

```
public String getQuestion() {  
    return question;  
}  
  
public void setQuestion(String question) {  
    this.question = question;  
}  
  
public List<AnswerTable> getAnswerList() {  
    return answerList;  
}  
  
public void setAnswerList(List<AnswerTable> answerList) {  
    this.answerList = answerList;  
}  
}
```

```
package com.telusko.model;  
  
import jakarta.persistence.CascadeType;  
import jakarta.persistence.Column;  
import jakarta.persistence.Entity;  
import jakarta.persistence.Id;  
import jakarta.persistence.ManyToOne;
```

```
@Entity  
public class AnswerTable  
{  
    @Id  
    @Column(name="answer_id")  
    private Integer id;
```

```
private String answer;
```

```
@ManyToOne(cascade=CascadeType.ALL)
```

```
private QuestionTable questionTable;
```

```
public QuestionTable getQuestion() {  
    return questionTable;  
}
```

```
public void setQuestionTable(QuestionTable questionTable) {  
    this.questionTable = questionTable;  
}
```

```
public AnswerTable()  
{  
    System.out.println("Zero param Constructor of Answer");  
}
```

```
public Integer getId() {  
    return id;  
}
```

```
public void setId(Integer id) {  
    this.id = id;  
}
```

```
public String getAnswer() {  
    return answer;  
}
```

```
public void setAnswer(String answer) {  
    this.answer = answer;  
}
```

```
    @Override
    public String toString() {
        return "AnswerTable [id=" + id + ", answer=" + answer + ",
questionTable=" + questionTable + "]";
    }
}
```

```
package com.telusko.app;

import java.util.ArrayList;
import java.util.List;

import org.hibernate.HibernateException;
import org.hibernate.Session;
import org.hibernate.SessionFactory;
import org.hibernate.Transaction;
import org.hibernate.cfg.Configuration;

import com.telusko.model.AnswerTable;
import com.telusko.model.QuestionTable;

public class LaunchStandardApp
{

    public static void main(String[] args)
    {
        Configuration config = null;
        SessionFactory sessionFactory = null;
        Session session = null;
        Transaction transaction = null;
    }
}
```

```
boolean flag=false;

config=new Configuration();

config.configure();

sessionFactory=config.buildSessionFactory();

session=sessionFactory.openSession();

QuestionTable q1=new QuestionTable();
q1.setId(1);
q1.setQuestion("What is Hibernate?");

AnswerTable answer1 =new AnswerTable();
answer1.setId(1);
answer1.setAnswer("Hibernate is an ORM Framewok");
answer1.setQuestionTable(q1);

AnswerTable answer2 =new AnswerTable();
answer2.setId(2);
answer2.setAnswer("Hibernate is an implementation of JPA");
answer2.setQuestionTable(q1);

List<AnswerTable> answers=new ArrayList<AnswerTable>();
answers.add(answer1);
answers.add(answer2);

q1.setAnswerList(answers);

//      QuestionTable question = session.get(QuestionTable.class,1);
//
//      System.out.println(question.getQuestion());
////    answer=question.getAnswerList();
```

```
//  
//      for(AnswerTable answers: question.getAnswerList())  
//      {  
//          System.out.println(answers.getAnswer());  
//      }
```

```
    try  
    {  
transaction=session.beginTransaction();
```

```
        session.persist(q1);
```

```
        flag=true;
```

```
    }  
    catch(HibernateException e)  
    {  
        e.printStackTrace();  
    }  
    catch(Exception e)  
    {  
        e.printStackTrace();  
    }
```

```
    finally  
    {  
if(flag==true)  
    {  
        transaction.commit();  
    }  
    else  
    {  
        transaction.rollback();
```

```
    }  
  
    session.close();  
    sessionFactory.close();  
  
    }  
  
    }  
  
}
```

Output:

```
mysql> use telusko_db;  
Database changed  
mysql> show tables;  
+-----+  
| Tables_in_telusko_db |  
+-----+  
| answertable           |  
| questiontable         |  
| questiontable_answertable |  
+-----+  
3 rows in set (0.02 sec)
```



```
mysql> select * from answertable;
+-----+-----+-----+
| answer_id | questionTable_question_id | answer |
+-----+-----+-----+
| 1 | 1 | Hibernate is an ORM Framewok |
| 2 | 1 | Hibernate is an implementation of JPA |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> select * from questiontable;
+-----+-----+
| question_id | question |
+-----+-----+
| 1 | What is Hibernate? |
+-----+-----+
1 row in set (0.00 sec)
```

INFO: HHH10001501: Connection obtained from JdbcConnectionAccess [org.hibernate.en]

Hibernate:

```
create table QuestionTable (
    question_id integer not null,
    question varchar(255),
    primary key (question_id)
) engine=InnoDB
```

Hibernate:

```
create table QuestionTable_AnswerTable (
    QuestionTable_question_id integer not null,
    answerList_answer_id integer not null
) engine=InnoDB
```

Hibernate:

```
alter table QuestionTable_AnswerTable
    add constraint UK_npi4sdg5u0tld2fpyi2ky0f1n unique (answerList_answer_id)
```

Hibernate:

```
alter table AnswerTable
    add constraint FKnpjhb8cv0jqswrceb2rwmtn2ik
    foreign key (questionTable_question_id)
    references QuestionTable (question_id)
```

Hibernate:

```
alter table QuestionTable_AnswerTable
    add constraint FK6511gddj5t07otv3ld2wj8ki5
    foreign key (answerList_answer_id)
    references AnswerTable (answer_id)
```

Hibernate:

```
alter table QuestionTable_AnswerTable
    add constraint FK1jck3814ug4p33ao7wx2iob5q
    foreign key (QuestionTable_question_id)
    references QuestionTable (question_id)
```

Zero Param Constructor of Question

Zero param Constructor of Answer

Zero param Constructor of Answer

```

Hibernate:
    insert
    into
        QuestionTable
        (question, question_id)
    values
        (?, ?)
Hibernate:
    insert
    into
        AnswerTable
        (answer, questionTable_question_id, answer_id)
    values
        (?, ?, ?)
Hibernate:
    insert
    into
        AnswerTable
        (answer, questionTable_question_id, answer_id)
    values
        (?, ?, ?)
Hibernate:
    insert
    into
        QuestionTable_AnswerTable
        (QuestionTable_question_id, answerList_answer_id)
    values
        (?, ?)
Hibernate:
    insert
    into
        QuestionTable_AnswerTable
        (QuestionTable_question_id, answerList_answer_id)
    values
        (?, ?)

```

- **Explanation:**

- The @OneToMany annotation defines a one-to-many relationship between QuestionTable and AnswerTable.
- CascadeType ALL ensures that operations like saving or deleting a QuestionTable object are cascaded to the related AnswerTable objects.
- You can use session.get(QuestionTable.class, id) to retrieve the QuestionTable object and associated answers.

- **Explanation:**

- @ManyToOne is used to map the relationship from AnswerTable to QuestionTable.
- Each AnswerTable instance corresponds to one QuestionTable entity.
-